

Networks

Supply and demand

Michel Bierlaire

Optimization: principles and algorithms



Supply and demand

Nodes can also be associated with quantities.

For instance, the supply is a quantity of flow that a node is injecting on a network.

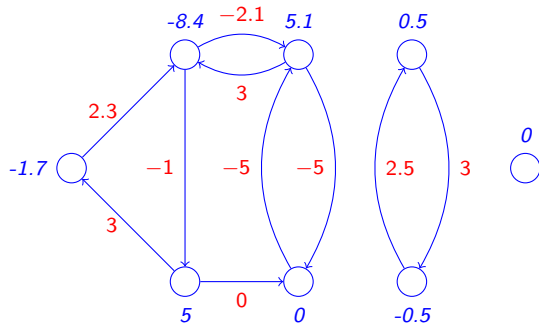
In logistics, a warehouse is supplying flow of goods on the network.

The demand is a quantity of flow absorbed by a node.

In logistics, a customer is collecting the flow from the network.

We characterize these notions in our mathematical formalism.

Divergences



Divergence

Flow leaving a node

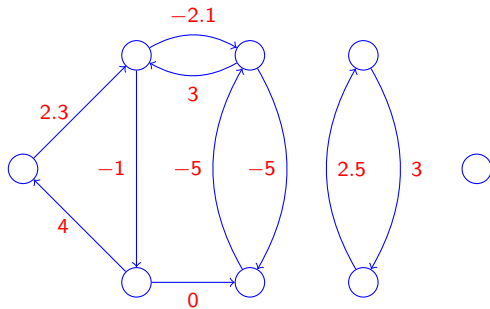
$$\sum_{j|(i,j) \in \mathcal{A}} x_{ij}.$$

$$-2.1 - 1 = -3.1$$

Flow entering a node

$$\sum_{k|(k,i) \in \mathcal{A}} x_{ki}.$$

$$2.3 + 3 = 5.3$$

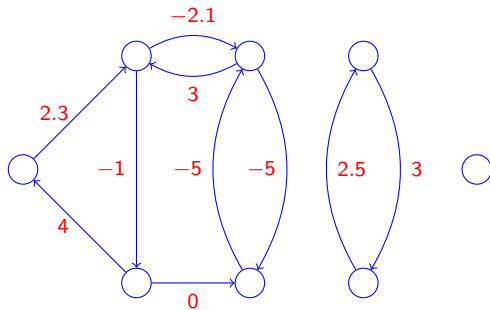


Divergence

Divergence

$$\text{div}(x)_i = \sum_{j|(i,j) \in \mathcal{A}} x_{ij} - \sum_{k|(k,i) \in \mathcal{A}} x_{ki}.$$

$$-3.1 - 5.3 = -8.4$$



Supply and demand

Supply node

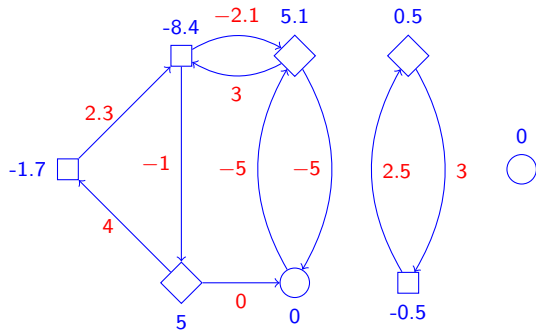
$$\text{div}(x)_i > 0.$$

Demand node

$$\text{div}(x)_i < 0.$$

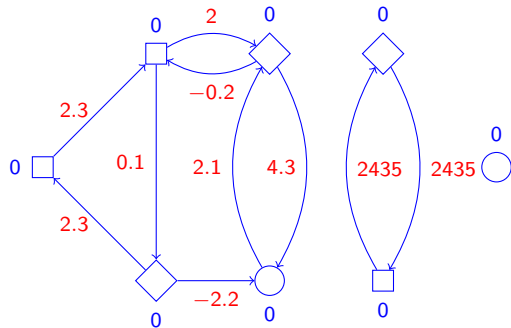
In any case

$$\sum_{i \in \mathcal{N}} \text{div}(x)_i = 0.$$



Circulation

$$\text{div}(x)_i = 0, \forall i \in \mathcal{N}.$$



Summary

The net quantity of flow at a node is called its divergence

If it is positive, it is a supply node

If negative, it is a demand node

If all divergences are zero, the flow is just circulating on the network.

Note that the sum of all divergences is always zero.